Vacancy-Induced Dzyaloshinskii–Moriya Interaction and Spin Canting in Monolayer Fe_3GeTe_2

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Fe_3GeTe_2, a van der Waals ferromagnet that retains magnetic order down to the monolayer, hosts Néel-type skyrmions and chiral domain walls in flakes, although the symmetry of its pristine bulk and monolayer structures forbids a net Dzyaloshinskii–Moriya interaction (DMI). Fe deficiency has been proposed to resolve this apparent contradiction, but how vacancies on the two inequivalent Fe sublattices cant the surrounding moments, and whether they can generate a net DMI, remains unknown. Using noncollinear density functional theory with spin–orbit coupling, we show that vacancies on both the outer-plane Fe(1) and central-plane Fe(2) sublattices induce local spin canting, reaching 0.95° and 0.44°, respectively. By comparing the canting for in-plane and out-of-plane magnetization, we find that the nearest-neighbor canting is predominantly antisymmetric, which points to DMI as its main origin. The two vacancy types differ in their symmetry. Fe(2) vacancies, which are more stable by 0.41 eV, preserve the horizontal mirror, so that their DMI contributions cancel between the outer planes. By symmetry, Fe(1) vacancies break this mirror and allow a net Néel-type DMI when distributed unequally between the outer planes. For an imbalance of 5% of the Fe(1) sites, we estimate from the spin–orbit torques that this DMI lies below the values inferred from experiment, which indicates that Fe vacancies alone are unlikely to produce a net DMI of the observed size.

Introduction—Néel-type skyrmions and chiral spin spirals are observed in flakes and lamellae of the van der Waals ferromagnet Fe_3GeTe_2 (FGT), tens of nanometers to micrometers thick [1–6]. Bloch-type bubbles, which require no chiral interaction, also occur [6, 7]. Yet the symmetry of both the stoichiometric bulk crystal [8] and the monolayer excludes a net Dzyaloshinskii–Moriya interaction (DMI), which usually stabilizes Néel-type textures. In the monolayer, individual Fe–Fe bonds carry DMI vectors [9, 10]. For the point group D_{3h}, however, the terms linear in the magnetization gradient either vanish or reduce to total derivatives, so that no Lifshitz invariant, and hence no net DMI, survives [9, 11, 12]. The DMI then acts only at boundaries, and canted moments are predicted at the edges of FGT flakes [9, 11]. The textures have been attributed to interfaces [1, 2] or to Fe deficiency, through unequal occupancy of the central Fe sites in alternate layers [5] or an off-plane displacement of these Fe atoms [6]. A recent conference abstract reports a twofold enhancement of the DMI when Fe vacancies are included in a first-principles defect Hamiltonian [13]. It remains open, however, how vacancies on the two Fe sublattices cant the moments and whether the DMI that they induce can add up to a net value.

FGT is a metallic ferromagnet with perpendicular anisotropy [14, 15] that remains ferromagnetic down to a monolayer [16, 17]. As in other two-dimensional magnets, its long-range order relies on this anisotropy [18–22]. The monolayer has two outer planes of Fe(1) atoms and a central Fe(2)–Ge plane (Fig. 1). A horizontal mirror plane σ_h through the central plane maps one outer plane onto the other. Bulk FGT is usually Fe deficient, and the Fe content changes its Curie temperature, anisotropy, magnetization, spin excitations, and transport [23–29]. Structural studies place the vacancies on the Fe(2) site [23, 24, 27], and scanning tunneling microscopy (STM) resolves individual Fe(2) vacancies in bulk crystals [30, 31]. Density functional theory (DFT) calculations find that vacancies prefer the Fe(2) site [32, 33], although the local-density approximation (LDA) reverses this preference [33], and that an Fe(2) vacancy reduces the moments of the neighboring Fe(1) atoms [32, 33].

Defects can generate DMI locally. In metallic spin glasses, nonmagnetic impurities induce it through spin–orbit scattering [34]. In a model ferromagnet, a vacancy removes the local inversion symmetry of the surrounding bonds, and the resulting DMI cants the neighboring moments into a vortex-like pattern [35]. In monolayer Fe_5GeTe_2, Fe vacancies mediate a reconstruction in which competing exchange interactions cant the moments strongly [36]. A vacancy in FGT is the point-defect analog of a flake edge. If the collinear ferromagnet is stable, it can tilt the neighboring moments only through spin–orbit coupling (SOC). Without SOC and without an applied field, the DFT energy is invariant under global spin rotations, and the transverse field on each moment of a collinear state vanishes. The canting around a vacancy therefore probes the SOC-derived interactions that the vacancy disturbs, with a pattern set by the symmetry that the vacancy preserves. An Fe(2) vacancy retains σ_h, whereas an Fe(1) vacancy breaks it.

Here we calculate this canting from first principles for both vacancies in monolayer FGT and separate its anti-

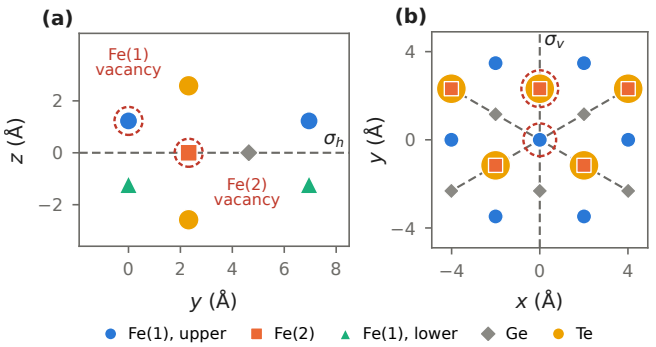


FIG. 1. Structure of monolayer FGT. (a) Atoms in the vertical mirror plane $x = 0$ through an Fe(1) column. The horizontal mirror plane σ_h contains the Fe(2) and Ge atoms and maps the upper Fe(1) plane onto the lower one. (b) Top view with the three vertical mirror planes σ_v through an Fe(1) column. Each Fe(1) symbol covers the lower Fe(1) atom of the same column, and each Fe(2) atom lies between two Te atoms. Dashed circles mark the Fe(1) and Fe(2) sites that are vacated, one in each calculation.

symmetric (DMI) and symmetric contributions by comparing two magnetization directions. We find that the canting next to both vacancies is predominantly antisymmetric (DMI-type). The Fe(2) vacancy, which is the stable one, yields no net DMI by symmetry. The Fe(1) vacancy allows a net Néel-type DMI, which we estimate from the SOC torques to be small.

Methods—We solve the Kohn–Sham equations [37] with the finite-element code DFT-FE [38, 39], using its noncollinear spinor formulation [40–42], the Perdew–Burke–Ernzerhof (PBE) functional [43], D4 dispersion corrections [44, 45], and fully relativistic optimized norm-conserving Vanderbilt pseudopotentials from the PseudoDojo stringent set [46–48]. We do not add a Hubbard U. For monolayer FGT, DFT combined with dynamical mean-field theory (DMFT) gives Fe moments within 15% of the values of the generalized-gradient approximation and the same easy axis, whereas DFT+U overestimates the moments [10]. For bulk FGT, DFT overestimates the ordered moment, and corrections beyond the generalized-gradient approximation have been argued to matter [31, 49]. Our pristine PBE moments are 2.43 μ_B for Fe(1) and 1.50 μ_B for Fe(2), close to the published 2.49 and 1.50 μ_B [10].

The rectangular 24.07×20.86 Å supercell of 36 formula units is periodic in plane, with z along the layer normal. Removing one Fe atom gives one vacancy per 108 Fe atoms, with the Fe(1) vacancy placed in the upper ($z > 0$) plane (Fig. 1). We use a Γ-centered $4 \times 4 \times 1$ k-point grid [50], an electronic temperature of 50 K, and seventh-order finite elements. The self-consistent-field (SCF) iterations combine Anderson mixing [51] with Resta preconditioning [52] and residual-based Chebyshev-filtered subspace iteration [53, 54]. Atomic positions were relaxed with SOC at fixed cell vectors, with the magnetization initialized along $+z$, the easy axis. For both vacancies and the pristine cell, we repeated the SCF calculation at the same geometry with the magnetization initialized along $+x$.

We define the moment of atom A as $\boldsymbol{\mu}_A = \int W_A(\mathbf{r})\, \mathbf{m}(\mathbf{r})\, d\mathbf{r}$, where $\mathbf{m}$ is the spin magnetization density and W_A is a spherical window of radius about 2 bohr around the nucleus. Including the orbital moments, about 0.055 μ_B per Fe, would raise the largest angles by about 20% [see the Supplemental Material (SM) [55]]. We express all moments in a frame in which the summed Fe moment points along the reference axis (z unless stated otherwise). The canting angle θ_A is the angle between $\boldsymbol{\mu}_A$ and z, and the transverse moment $\boldsymbol{\mu}_{A,\perp}$ is the component of $\boldsymbol{\mu}_A$ perpendicular to z. We resolve $\boldsymbol{\mu}_{A,\perp}$ into radial and tangential components relative to the in-plane position of the vacancy, with positive radial components pointing away from the vacancy. Further details and numerical checks are given in the SM.

Canting around the vacancies—The Fe(2) vacancy is 0.41 eV lower in energy than the Fe(1) vacancy. PBE calculations in a smaller cell found the same preference, with 0.18 eV [33]. As in published calculations [32, 33], its nearest Fe(1) neighbors lose moment (2.43 to 2.11 μ_B) and its nearest Fe(2) neighbors gain moment (1.50 to 1.67 μ_B). Around the Fe(1) vacancy, the nearest Fe(2) neighbors instead lose moment, from 1.50 to 1.23 μ_B, as also found in Ref. [33].

Both vacancies also cant the surrounding moments (Fig. 2). The largest angle is 0.95° for the Fe(1) vacancy and 0.44° for the Fe(2) vacancy, compared with 0.014° in the pristine cell. The pristine lattice has a threefold axis through each vacant site, but neither the rectangular cell nor the k-point grid has this axis. For the shells within 5 Å, sites related by it differ in angle by up to 21%. Farther out the differences are larger, because the periodic images lie at different distances along x and y. We use the spread of the near shells as an estimate of the numerical uncertainty of the angles, not as a convergence bound (SM). The differences discussed below exceed it.

Around the Fe(2) vacancy, the canting is largest on the six nearest Fe(1) neighbors at 2.62 Å, three in each outer plane. For magnetization along $+z$, their transverse moments point radially toward the vacancy in the upper plane and away from it in the lower plane. The central-plane Fe(2) neighbors remain uncanted ($\theta_A \leq 0.007°$), whereas around the Fe(1) vacancy all three planes are canted. Its three nearest Fe(2) neighbors are canted radially toward the vacancy by 0.79–0.95°. The Fe(1) neighbors in the same plane, at 4.0 Å, are canted by 0.53–0.62°, and those in the opposite (lower) plane, at 4.7 Å, by 0.76–0.90° [Fig. 2(c)]. These Fe(1) atoms are canted mostly tangentially, with radial components accounting for only 18% (upper plane) and 38% (lower plane) of their

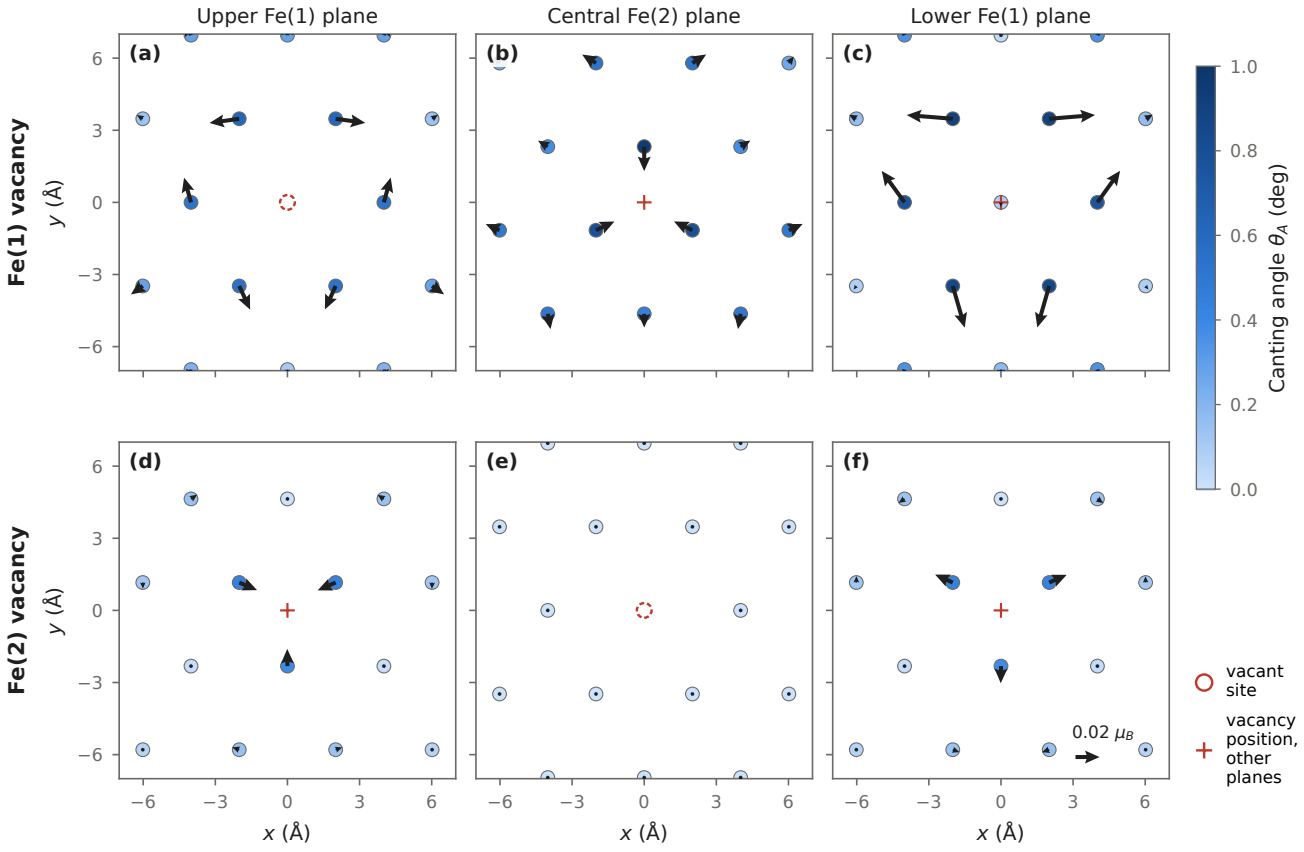


FIG. 2. Transverse moments $\boldsymbol{\mu}_{A,\perp}$ (arrows) and canting angles θ_A (color) in the three Fe planes around the Fe(1) vacancy (top row) and the Fe(2) vacancy (bottom row). The reference arrow in (f) corresponds to 0.02 μ_B. Atoms are drawn at their pristine positions. The dashed circle marks the vacant site and the cross its in-plane position in the other planes.

summed squared transverse moments.

Symmetry constraints: With the magnetization along z, the horizontal mirror σ_h maps an axial vector as $(m_x, m_y, m_z) \mapsto (-m_x, -m_y, m_z)$. Because the Fe(2) vacancy retains σ_h, the transverse moments of the two outer planes must be opposite and those of the central plane must vanish. A vertical mirror σ_v combined with time reversal reverses only the in-plane component normal to the mirror, so an atom on a σ_v plane through the vacancy can cant only radially. The three such planes through each vacancy site contain the nearest neighbors, whose squared transverse moments are 99% radial (SM). Symmetry thus fixes the canting axis of the nearest Fe(2)-vacancy neighbors and the relative sign of the two outer planes.

Because the Fe(1) vacancy breaks σ_h, symmetry neither relates the two outer planes nor forbids canting in the central plane. Its Fe(1) neighbors at 4.0 and 4.7 Å also lie between the σ_v planes, where tangential components are allowed.

Because $\tan\theta_A = |\boldsymbol{\mu}_{A,\perp}|/\mu_{A,z}$, the largest angle partly reflects the reduced moment (1.23 μ_B) of the Fe(2) neighbors of the Fe(1) vacancy. The largest transverse moments, 0.032–0.038 μ_B, are instead on lower-plane Fe(1) atoms with nearly unchanged moments [Fig. 3(c)].

The canting does not decay to zero within the cell [Fig. 3(a,b)]. Beyond 6 Å, angles still reach 0.23° around the Fe(2) vacancy and 0.53° around the Fe(1) vacancy. All results, including those near the vacancies, therefore refer to this periodic array rather than to an isolated vacancy.

Origin of the canting—We expand the magnetic energy for small transverse deviations $\boldsymbol{\delta}_i$ of the unit moment directions from a collinear reference,

$$E = E_0 + \sum_i \mathbf{g}_i \cdot \boldsymbol{\delta}_i + \tfrac{1}{2} \sum_{ij} \boldsymbol{\delta}_i \cdot \mathcal{K}_{ij} \boldsymbol{\delta}_j . \qquad (1)$$

The canting is then the linear response $\sum_j \mathcal{K}_{ij}\boldsymbol{\delta}_j = -\mathbf{g}_i$ [56, 57]. Isotropic exchange contributes to the stiffness $\mathcal{K}$ but not to the transverse field $\mathbf{g}_i$, which requires SOC. In the pristine layer, every Fe site lies on a threefold axis, which forces $\mathbf{g}_i$ to vanish. A vacancy lowers the symmetry of the surrounding sites and lifts this constraint.

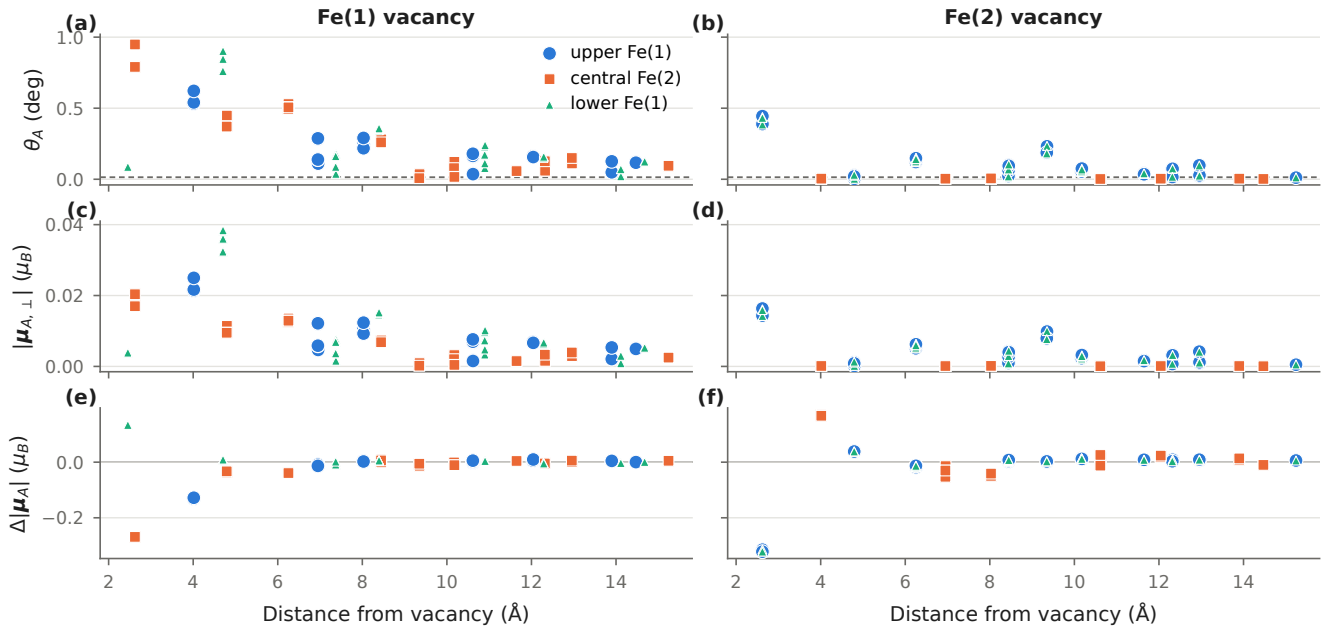


FIG. 3. Canting angle θ_A (a,b), transverse moment $|\boldsymbol{\mu}_{A,\perp}|$ (c,d), and change in moment magnitude relative to the same atom in the pristine cell (e,f) for every Fe atom, as a function of distance from the vacancy. The dashed line in (a,b) marks the largest pristine canting angle. For the Fe(2) vacancy, upper- and lower-plane values coincide by symmetry. Distances use pristine positions and minimum periodic images.

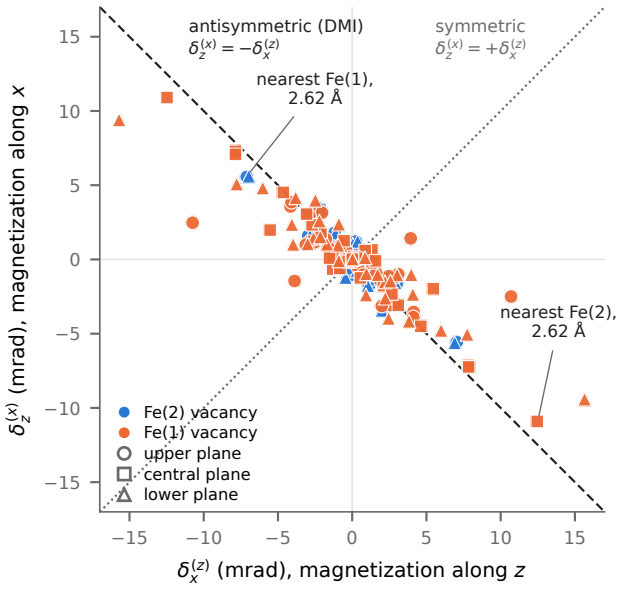


FIG. 4. Canting component $\delta_z^{(x)}$ of each Fe atom in the state magnetized along x against $\delta_x^{(z)}$ in the state magnetized along z, around the Fe(2) (blue) and Fe(1) (orange) vacancies. Antisymmetric (DMI-type) responses lie on the dashed line and symmetric ones on the dotted line. Atoms fixed at the origin by symmetry are omitted.

For example, a bond with DMI vector $\mathbf{D}_{ij}$ and energy $\mathbf{D}_{ij} \cdot (\mathbf{e}_i \times \mathbf{e}_j)$ [58, 59] contributes $\hat{\mathbf{z}} \times \mathbf{D}_{ij}$ to $\mathbf{g}_i$, where $\mathbf{e}_i$ is the unit moment direction. These contributions sum to zero on each pristine site, so removing atom 0 changes $\mathbf{g}_i$ by

$$\Delta\mathbf{g}_i = -\hat{\mathbf{z}} \times \mathbf{D}_{i0}, \tag{2}$$

if the other bonds are unchanged. The DMI acts in the same way at the edge of a flake [9, 11] and around a vacancy in a model ferromagnet [35].

When a mirror plane contains both atoms of a bond, $\mathbf{D}_{i0}$ is perpendicular to that plane [59]. For a bond in a vertical mirror plane through the vacancy, $\mathbf{D}_{i0}$ is therefore azimuthal and Eq. (2) gives a radial $\Delta\mathbf{g}_i$, as observed for the nearest neighbors of both vacancies. The Fe(1) neighbors of the Fe(1) vacancy lie off these planes. For the same-plane neighbors at 4.0 Å, however, a mirror plane bisects the bond and confines $\mathbf{D}_{i0}$ to that plane. Equation (2) then again gives only a radial $\Delta\mathbf{g}_i$, so their tangential canting must come from changes in the remaining interactions, from symmetric terms, or from the response to fields on other atoms. For the lower-plane neighbors at 4.7 Å, a twofold axis perpendicular to the bond makes $\Delta\mathbf{g}_i$ purely tangential. For the Fe(2) vacancy, σ_h maps each bond in one outer plane onto one in the other and reverses the in-plane components of its DMI vector, consistent with the opposite canting of the two planes.

Antisymmetric and symmetric contributions: Symmetric anisotropic terms, such as a locally tilted single-site anisotropy, obey the same symmetry constraints,

so this agreement does not by itself identify the DMI. First-principles studies separate the two through the relativistic exchange tensor [60–63], through energy mapping [64, 65], or through spin spirals of opposite chirality [66, 67]. We instead compare the self-consistent canting for two magnetization directions, which probes the first-order term of Eq. (1), to which the DMI contributes [68]. This comparison needs only the two states that also give the anisotropy energy. We write the SOC-derived couplings as $E = \sum_{ij} \mathbf{e}_i \cdot \Gamma_{ij} \mathbf{e}_j$, with single-site terms included as $i = j$. For a reference along z, the transverse field on atom i along x is then proportional to $\sum_j (\Gamma_{ij})_{xz}$, and for a reference along x, the field along z is proportional to $\sum_j (\Gamma_{ij})_{zx}$. The symmetric part of Γ_{ij} therefore gives $\delta_{i,z}^{(x)} = +\delta_{i,x}^{(z)}$, and the antisymmetric part, the DMI, gives $\delta_{i,z}^{(x)} = -\delta_{i,x}^{(z)}$, where the superscript denotes the reference direction. The combinations $\delta_i^{\mathrm{s}} = (\delta_{i,x}^{(z)} + \delta_{i,z}^{(x)})/2$ and $\delta_i^{\mathrm{a}} = (\delta_{i,x}^{(z)} - \delta_{i,z}^{(x)})/2$ then give the symmetric and antisymmetric contributions atom by atom. These relations assume bilinear couplings that do not depend on the magnetization direction. They also assume that the stiffness $\mathcal{K}$ is the same in both states and does not mix the compared components with the y components (SM). Deviations enter both combinations, so the split identifies an antisymmetric, DMI-type response rather than the coupling tensor itself. Fourth-order interactions, which this point group allows and which have been reported for monolayer FGT [12, 69], are not separated by this comparison.

The states magnetized along x lie higher in energy. The anisotropy energy, the electronic free-energy difference $E_x - E_z$ at 50 K, is 0.96 meV per Fe in the pristine cell, within the range of 0.9 to 1.3 meV per Fe published for the monolayer [9, 10, 15]. On our k-point grid, the Fe(2) vacancy lowers the anisotropy energy of the cell by 12 meV and the Fe(1) vacancy raises it by 8 meV, whereas PBE calculations in a smaller cell found small increases for both [33]. We have not tested these small changes on a denser grid (SM).

Figure 4 compares the canting in the two directions. For the four nearest Fe(1) neighbors of the Fe(2) vacancy off the σ_v plane normal to x, $\delta_z^{(x)}$ is -0.78 to -0.81 times $\delta_x^{(z)}$. Their antisymmetric part, $|\delta_i^{\mathrm{a}}| = 6.3$ mrad, accounts for 98.7% of the squared canting, and the symmetric part amounts to 11% of it in amplitude. For the two nearest Fe(2) neighbors of the Fe(1) vacancy off this plane, the ratio is -0.87. Their antisymmetric part, $|\delta_i^{\mathrm{a}}| = 11.7$ mrad, accounts for 99.6%, and the symmetric part amounts to 7% of it in amplitude. A difference in stiffness between the two states can change the magnitude of the response but not its sign. Around the Fe(1) vacancy, the antisymmetric part also dominates farther out within this periodic cell, with 94% for the lower-plane Fe(1) atoms at 4.7 Å and 88–99% in each distance range beyond 5.5 Å (SM). Two shells do not follow this pattern. The second Fe(1) shell of the Fe(2) vacancy cants by at most 0.03°, which is comparable to the components that the threefold symmetry forbids, and we do not classify it. The tangential canting of the same-plane Fe(1) neighbors of the Fe(1) vacancy, at 4.0 Å, is about equally antisymmetric and symmetric (SM).

Within these assumptions, the canting next to both vacancies arises predominantly from DMI contributions to $\mathbf{g}_i$. These contributions cancel in the pristine layer, and the vacancy removes the cancellation, in agreement with Eq. (2). The comparison cannot, however, separate removed from modified bonds. It probes only the xz element of Γ_{ij}, that is, the y components of the DMI vectors, and it does not test the atoms on the σ_v plane normal to x.

The SOC torques give a second test, which does not use the equal-stiffness assumption (SM). The torque that the SOC of an Fe atom exerts on its own spin gives the transverse field from that SOC alone. Between the z and x states, its DMI-type part keeps its sign and its symmetric part reverses. At the nearest neighbors, where the fields are largest, the DMI-type part accounts for 86% [Fe(2) vacancy] and 92% [Fe(1) vacancy] of the squared torque, in line with the canting. These DMI-type fields are 0.2 and 0.4 meV per radian. In the second shell the torque is mostly symmetric. Farther out the two tests give different fractions (SM), and the DMI-type torque from Fe SOC is confined to the nearest one or two shells. The torque omits the fields from the SOC of all other atoms, which reach an Fe atom through the spin current into its atomic region. The Te torques are mostly symmetric. Their DMI-type part is nevertheless comparable to that of Fe within 5 Å and extends farther, so the SOC of both species contributes to the DMI-type torque pattern (SM).

Local and net DMI: Whether these local contributions add up to a net DMI of the layer is a question of symmetry. Because the Fe(2) vacancy preserves σ_h, the Néel-type contributions of its two outer planes are opposite and cancel (Table S15 in the SM). A layer with a statistically threefold-symmetric distribution of such vacancies keeps D_{3h}, which excludes a net DMI. The same argument applies to the bulk crystal. Unequal occupancy of the Fe(2) sites in alternate layers [5] lowers the space group from $P6_3/mmc$ to $P\bar{6}m2$, whose point group is again D_{3h} (SM). This imbalance alone therefore allows no net DMI, and the polar structures reported for Fe-deficient crystals [5, 6] require a further distortion, such as Fe atoms in the van der Waals gap [5] or an off-plane displacement of the Fe(2) atoms [6]. Such a displacement would break σ_h in the monolayer as well. Our relaxed Fe(2)-vacancy cell retains σ_h, but the mirror was present in every recorded geometry of the relaxation, which therefore does not test this instability (SM).

Around the Fe(1) vacancy, the radial antisymmetric

canting is inward on the nearest central-plane neighbors and outward on the Fe(1) neighbors of both outer planes, with no mirror partner (SM). Unequal vacancy densities in the upper and lower Fe(1) planes lower the statistical symmetry to C_{3v}, which allows a net Néel-type DMI. In the bulk crystal, this requires the same sense of the imbalance in every layer.

The SOC torques give an estimate of its size. For a vacancy in a layer without net DMI, the energy of a long-wavelength cycloidal spin spiral that is linear in its wave vector is given by the first moment of the DMI-type torques about the vacancy [70] (SM). For the Fe(2) vacancy, the contributions of the two outer planes to this moment cancel to within 0.1%. For the Fe(1) vacancy, the moment has the same sign for every cutoff radius. It amounts to 1.3 to 2.9 meV Å per vacancy for cutoffs between 5 and 9.5 Å and to 6.1 meV Å for the whole cell, where it is dominated by atoms near the cell boundary. For an excess of Fe(1) vacancies in one outer plane that amounts to 5% of its sites, these values correspond to a DMI constant of 0.01 to 0.04 mJ/m^2. Experiments give lower bounds of 0.1 to 0.2 mJ/m^2 for flakes [3, 4] and a value of 1.0 mJ/m^2 for a WTe_2/FGT heterostructure [1]. Our estimate is accurate only to its order of magnitude. It assigns each torque to the position of its atom, it is exact only to first order in SOC [71, 72], and it uses two states whose canting was relaxed separately (SM). Bond-resolved exchange tensors or the energy of long-wavelength spirals, which we have not computed, are needed to confirm it.

The energetics also disfavor this route. Within PBE, Fe(1) vacancies are 0.41 eV less stable than Fe(2) vacancies. At thermal equilibrium at 1000 K, this difference corresponds to a ratio of Fe(1) to Fe(2) vacancies of the order of 10^{-2}. A layer with a sizable excess of Fe(1) vacancies in one plane therefore requires growth or interface conditions that favor them.

Conclusions—Fe vacancies in monolayer FGT cant the surrounding Fe spin moments by up to about 1° through SOC, in a sublattice-dependent pattern. For both vacancies, the comparison of two magnetization directions and the SOC torques indicate that the canting next to the vacancy comes predominantly from DMI contributions that cancel in the pristine layer. Symmetry decides whether these local contributions add up. The Fe(2) vacancy, which is the stable one and the one found in structural studies, preserves σ_h. Its DMI-driven canting is opposite in the two outer planes, and it yields no net DMI. For the bulk crystal, the same argument shows that unequal Fe(2) occupancy between layers does not suffice either. The Fe(1) vacancy breaks σ_h, and Fe(1) vacancies distributed unequally between the outer planes allow a net Néel-type DMI. Our estimate from the SOC torques places it below the values inferred from experiment for an imbalance of a few percent. Within PBE, and for a layer that retains σ_h around its Fe(2) vacancies, Fe vacancies alone are therefore unlikely to produce a net DMI of the observed size. This points to distortions that break σ_h, such as an off-plane displacement of Fe(2) atoms or Fe in the van der Waals gap, and to interfaces.

These results are obtained within PBE, in a single cell, and on a single k-point grid. DMFT halves the strongest exchange coupling of the monolayer and raises its anisotropy energy [10], which would change both the stiffness and the fields that set the canting. Spin-polarized STM, which resolves noncollinear spin states of atomic chains and atom pairs whose chirality is set by the DMI [73–75], could test these predictions if it resolves in-plane spin asymmetries of a few percent near a vacancy (SM). On epitaxial FGT monolayers [76], the substrate breaks σ_h, which must be taken into account.

Data availability—The data supporting this study are available from the authors upon reasonable request.

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